Lecture 26

STAT 109: Introductory Biostatistics

Lecture 26: Two numerical variables — scatterplots and describing relationships

Learning objectives

- Make a **scatterplot** in **ggplot2** for two **numerical** variables (`aes(x = …, y = …)` and `geom_point()`).
- Describe a relationship in words using **form** (e.g. linear vs curved), **direction** (positive, negative, or none), and **strength** (strong, moderate, weak, or none).

Why two numerical variables together

When both variables are **numerical**, we often ask how one changes with the other: does taller vegetation go with wetter soil? Does leaf nitrogen track soil nitrate? A **scatterplot** puts one variable on the **horizontal** axis and one on the **vertical** axis; each point is one observational unit (a tree, a nest, a transect).

Scatterplots in ggplot2

Load data into a data frame (or tibble) with **two numeric columns**. Map them in `aes()` and add points.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)
```

Minimal pattern:

```
ggplot(your_dataset, aes(x = variable_on_x, y = variable_on_y)) +
  geom_point() +
  labs(title = "…", x = "…", y = "…")
```

Use `labs()` for titles and axis labels with **units** when you have them.

Describing a scatterplot in words

Read the cloud of points as a whole.

Form (shape). Is the trend **roughly a straight line** (**linear**), **bent** (**curved** / **nonlinear**), **split into clusters**, or **no clear pattern**?

Direction. As x increases, does y tend to **increase** (**positive** association), **decrease** (**negative** association), or show **no consistent** up-or-down trend?

Strength. How tightly do points follow the trend in the form? **Strong:** points closely hug a line or curve. **Moderate:**

signal is visible but some random scatter above and below is noticeable. **Weak:** only a faint drift. **None:** no useful trend (uniform random scatter around a horizontal band).

These labels are **judgment calls** in practice; the point is to communicate what you see before moving to formal models (later in the course).

Examples (biology and field settings)

The plots below use **made-up** data for teaching. Each chunk builds a small tibble, shows the **first 10 rows** with **`head(..., 10)`**, then plots all rows.

Strong positive, roughly linear

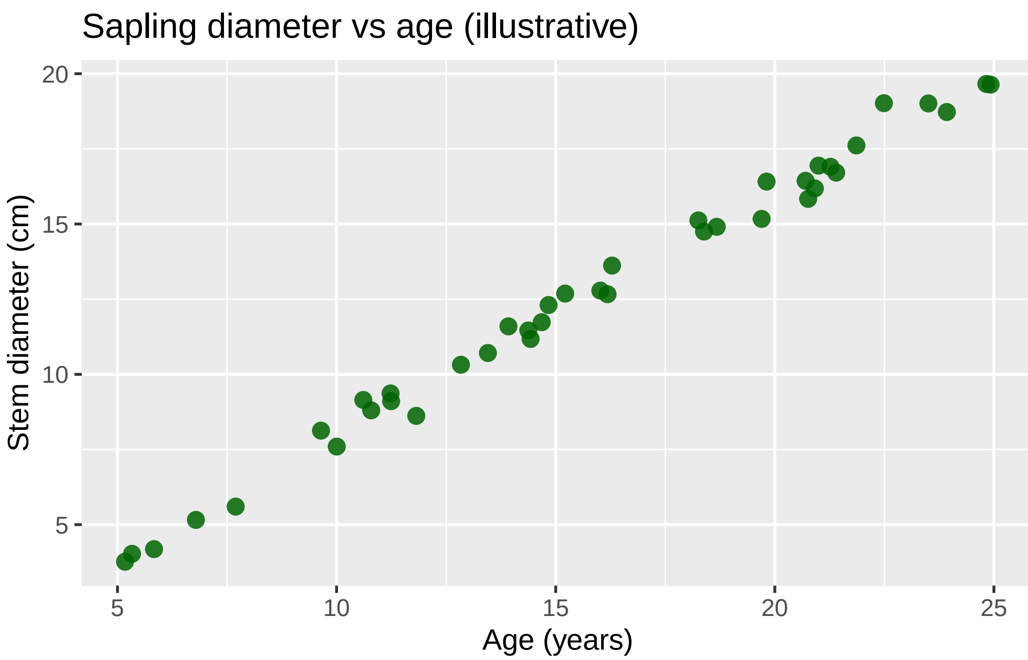
Context: Saplings in a plantation. **Stem diameter** (cm at breast height) vs **age** (years). Older stems are thicker; points fall near a straight line with little scatter.

```
set.seed(26)
trees_age_diam <- tibble(
  age_years = runif(40, 5, 25),
  diameter_cm = 0.8 * age_years + rnorm(40, 0, 0.4)
)

head(trees_age_diam, 10)
```

```
# A tibble: 10 x 2
   age_years diameter_cm
       <dbl>       <dbl>
 1      5.33        4.03
 2     10.8         8.80
 3     22.5        19.0
 4     21.0        16.9
 5     11.2         9.11
 6     14.4        11.2
 7     20.9        16.2
 8     19.7        15.2
 9     11.8         8.62
10     18.4        14.8
```

```
ggplot(trees_age_diam, aes(x = age_years, y = diameter_cm)) +
  geom_point(alpha = 0.85, size = 2.5, color = "darkgreen") +
  labs(
    title = "Sapling diameter vs age (illustrative)",
    x = "Age (years)",
    y = "Stem diameter (cm)"
  )
```



Words: Positive direction, **strong** strength, **linear** form.

Strong negative, roughly linear

Context: Stream sites along a mountain gradient. **Elevation** (m) vs **water temperature** (°C) at midday in summer. Higher sites tend to be cooler.

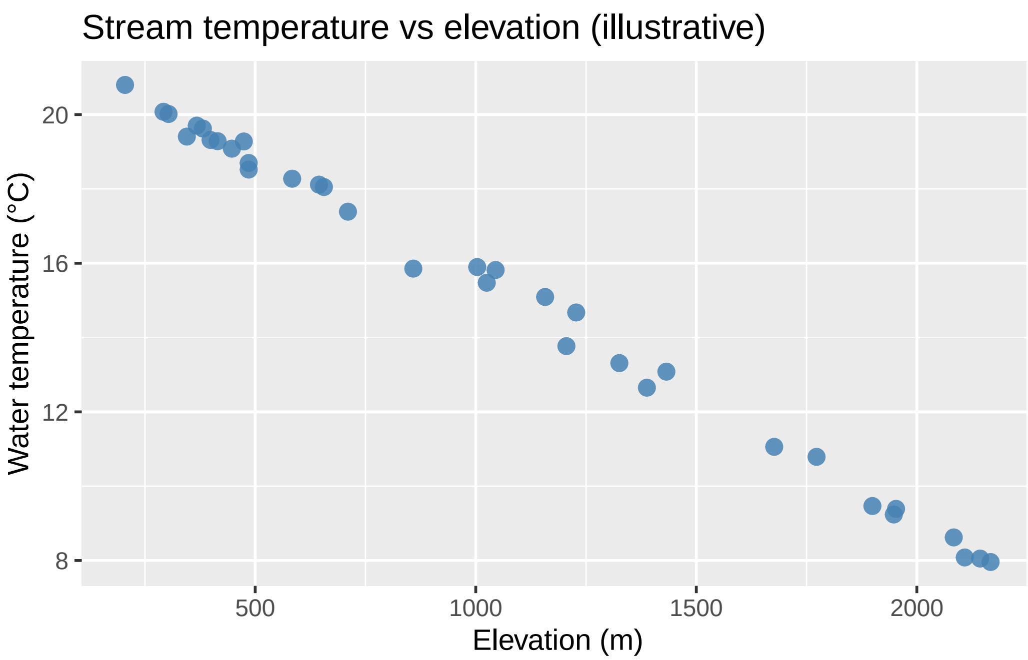
```
set.seed(27)
stream_temp <- tibble(
  elevation_m = runif(35, 200, 2200),
  water_temp_c = 22 - 0.0065 * elevation_m + rnorm(35, 0, 0.35)
)

head(stream_temp, 10)
```

```
# A tibble: 10 x 2
   elevation_m water_temp_c
         <dbl>        <dbl>
 1       2144.         8.05
 2        368.        19.7
 3       1948.         9.24
 4        858.        15.9
 5        645.        18.1
 6       1003.        15.9
 7        345.        19.4
 8        205.        20.8
 9        474.        19.3
10        584.        18.3
```

```
ggplot(stream_temp, aes(x = elevation_m, y = water_temp_c)) +
  geom_point(alpha = 0.85, size = 2.5, color = "steelblue") +
  labs(
```

```
    title = "Stream temperature vs elevation (illustrative)",
    x = "Elevation (m)",
    y = "Water temperature (°C)"
  )
```



Words: Negative direction, **strong** strength, **linear** form.

Moderate positive, linear

Context: Herbaceous plants along a transect. **Soil nitrate** (mg/kg) vs **leaf nitrogen** (%). Higher soil N tends to go with higher leaf N, with noticeable scatter from other factors.

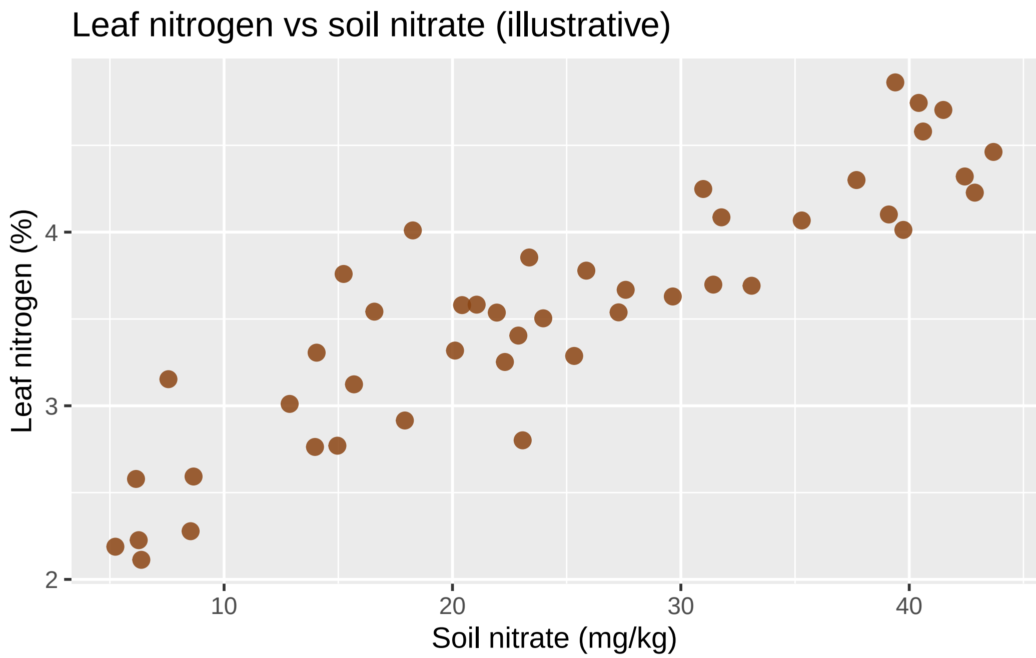
```
set.seed(28)
leaf_soil <- tibble(
  soil_nitrate_mgkg = runif(45, 5, 45),
  leaf_nitrogen_pct = 2.0 + 0.06 * soil_nitrate_mgkg + rnorm(45, 0, 0.35)
)

head(leaf_soil, 10)
```

```
# A tibble: 10 x 2
   soil_nitrate_mgkg leaf_nitrogen_pct
               <dbl>             <dbl>
 1              6.14              2.58
 2              8.53              2.28
 3             24.0               3.50
 4             40.4               4.74
 5              8.66              2.59
 6             35.3               4.07
 7              6.38              2.11
 8             39.7               4.01
 9             27.6               3.67
```

```
10              42.9              4.23
```

```r
ggplot(leaf_soil, aes(x = soil_nitrate_mgkg, y = leaf_nitrogen_pct)) +
  geom_point(alpha = 0.85, size = 2.5, color = "chocolate4") +
  labs(
    title = "Leaf nitrogen vs soil nitrate (illustrative)",
    x = "Soil nitrate (mg/kg)",
    y = "Leaf nitrogen (%)"
  )
```



Words: Positive direction, **moderate** strength, **linear** form.

Weak positive

Context: Forest plots. **Canopy openness** (percent sky visible) vs **understory seedling count** per m^2. Slight tendency for more seedlings where light is higher, but noise dominates.

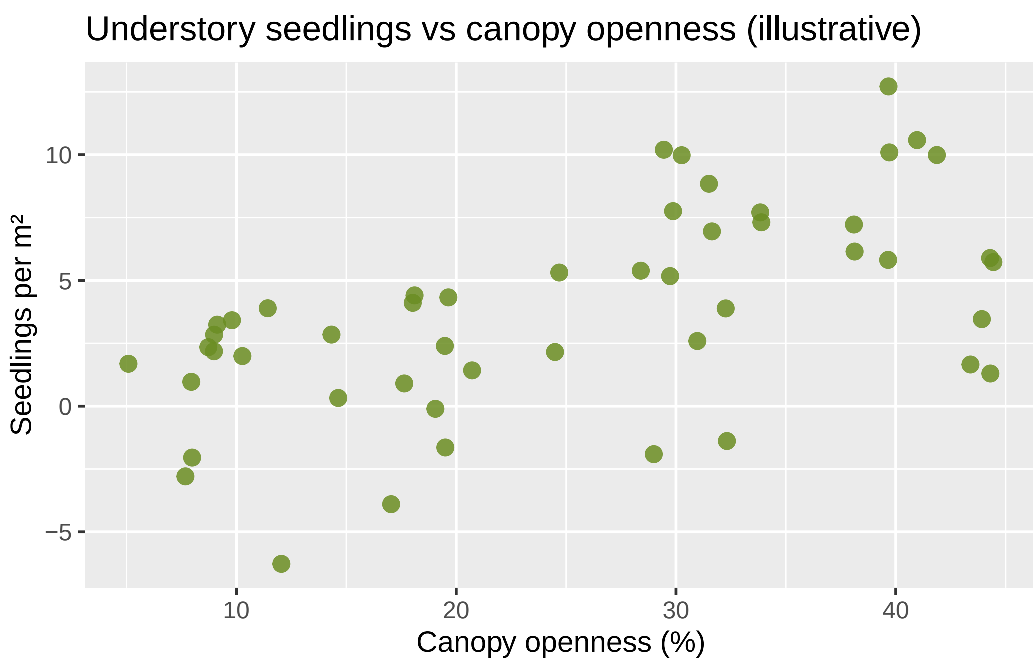
```r
set.seed(29)
understory <- tibble(
  canopy_open_pct = runif(50, 5, 45),
  seedlings_per_m2 = 2.5 + 0.08 * canopy_open_pct + rnorm(50, 0, 4)
)

head(understory, 10)
```

```
# A tibble: 10 x 2
   canopy_open_pct seedlings_per_m2
             <dbl>            <dbl>
 1            8.99            2.84
 2           14.6             0.325
 3            9.13            3.24
 4           18.0             4.11
 5           28.4             5.39
```

```
 6           8.72          2.35
 7          38.1           6.15
 8          39.7           5.82
 9           9.80          3.42
10          14.3           2.85
```

```
ggplot(understory, aes(x = canopy_open_pct, y = seedlings_per_m2)) +
  geom_point(alpha = 0.85, size = 2.5, color = "olivedrab") +
  labs(
    title = "Understory seedlings vs canopy openness (illustrative)",
    x = "Canopy openness (%)",
    y = "Seedlings per m²"
  )
```



Words: Positive direction, **weak** strength, **linear** drift at best.

No clear association

Context: Moose pellet counts along transects vs **soil pH**. No story linking these two in this toy dataset; points fill the rectangle.

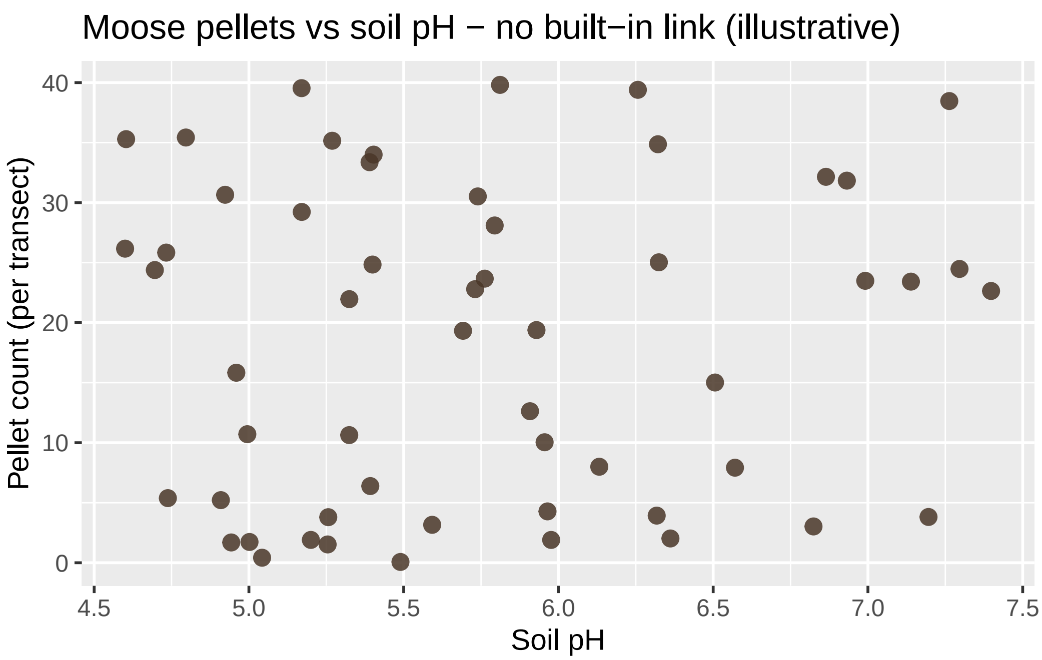
```
set.seed(30)
moose_ph <- tibble(
  soil_ph = runif(55, 4.5, 7.5),
  pellet_count = runif(55, 0, 40)
)

head(moose_ph, 10)
```

```
# A tibble: 10 x 2
   soil_ph pellet_count
     <dbl>        <dbl>
 1    4.80         35.4
```

```
 2    5.96         4.28
 3    5.59         3.16
 4    5.76        23.7
 5    5.40        34.0
 6    4.94         1.69
 7    7.20         3.81
 8    5.17        29.2
 9    7.40        22.6
10    4.92        30.7
```

```
ggplot(moose_ph, aes(x = soil_ph, y = pellet_count)) +
  geom_point(alpha = 0.85, size = 2.5, color = "#4a3728") +
  labs(
    title = "Moose pellets vs soil pH - no built-in link (illustrative)",
    x = "Soil pH",
    y = "Pellet count (per transect)"
  )
```



Words: No clear direction, **no** real strength, **no** simple form.

Nonlinear (curved): positive, strong

Context: Tree height (m) vs **age** (years) in even-aged stands. Growth is **fast** when young and **levels off** as crowns compete (diminishing returns), so the mean trend bends.

```
set.seed(31)
tree_curve <- tibble(
  age_years = runif(42, 8, 80),
  height_m = 25 * (1 - exp(-0.045 * age_years)) + rnorm(42, 0, 1.1)
)

head(tree_curve, 10)
```

```
# A tibble: 10 x 2
   age_years height_m
       <dbl>    <dbl>
 1      45.6     21.6
 2      76.4     24.9
 3      38.7     21.5
 4      38.6     20.5
 5      76.0     24.3
 6      40.6     23.5
 7      68.0     23.8
 8      20.4     14.8
 9      75.2     24.1
10      25.5     15.6
```

```r
ggplot(tree_curve, aes(x = age_years, y = height_m)) +
  geom_point(alpha = 0.85, size = 2.5, color = "forestgreen") +
  labs(
    title = "Tree height vs age - saturating growth (illustrative)",
    x = "Age (years)",
    y = "Height (m)"
  )
```

Words: Overall **positive** association, **strong** trend, **nonlinear** (curved) form.

Nonlinear: hump-shaped (moderate)

Context: Pollinator visits per flower vs **nectar volume** (μL). Very little nectar attracts few visitors; **moderate** nectar is best; **very high** volume may correlate with different flower types or confounding, producing a **hump** in this toy example.

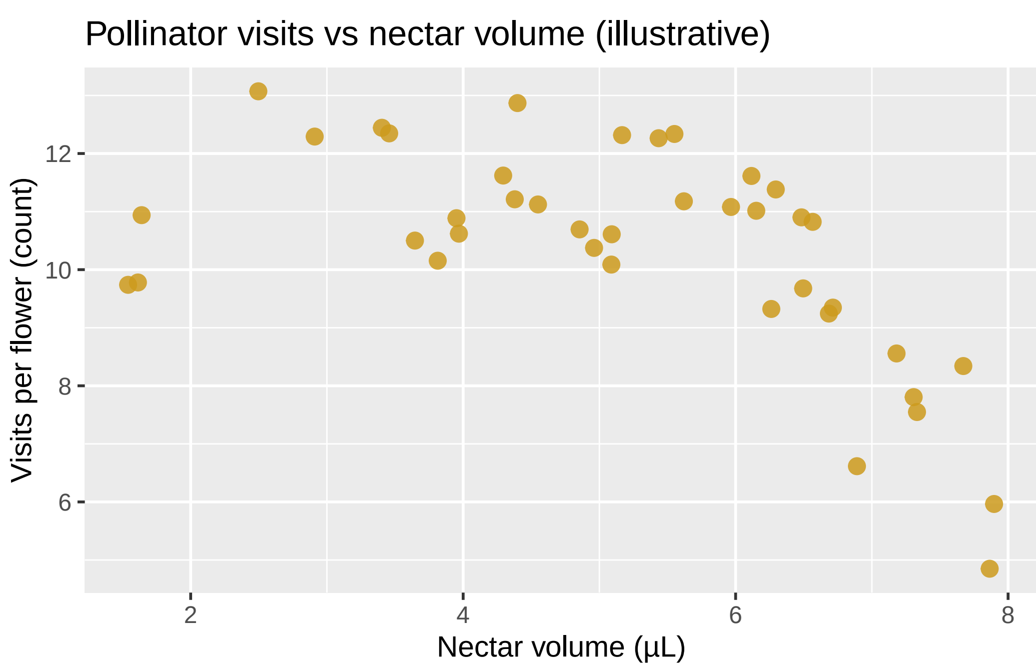
```r
set.seed(32)
nectar <- tibble(
```

```
  nectar_ul = runif(40, 0.5, 8),
  visits = -0.35 * (nectar_ul - 4)^2 + 12 + rnorm(40, 0, 1.2)
)

head(nectar, 10)
```

```
# A tibble: 10 x 2
   nectar_ul visits
       <dbl>  <dbl>
 1      4.29  11.6
 2      4.96  10.4
 3      6.57  10.8
 4      5.97  11.1
 5      1.64  10.9
 6      7.67   8.34
 7      6.15  11.0
 8      6.89   6.62
 9      5.55  12.3
10      3.40  12.4
```

```
ggplot(nectar, aes(x = nectar_ul, y = visits)) +
  geom_point(alpha = 0.85, size = 2.5, color = "goldenrod3") +
  labs(
    title = "Pollinator visits vs nectar volume (illustrative)",
    x = "Nectar volume (µL)",
    y = "Visits per flower (count)"
  )
```



Words: Nonlinear form (hump), **moderate** strength; direction is **not** simply positive or negative over the whole x range.

Strong negative, nonlinear

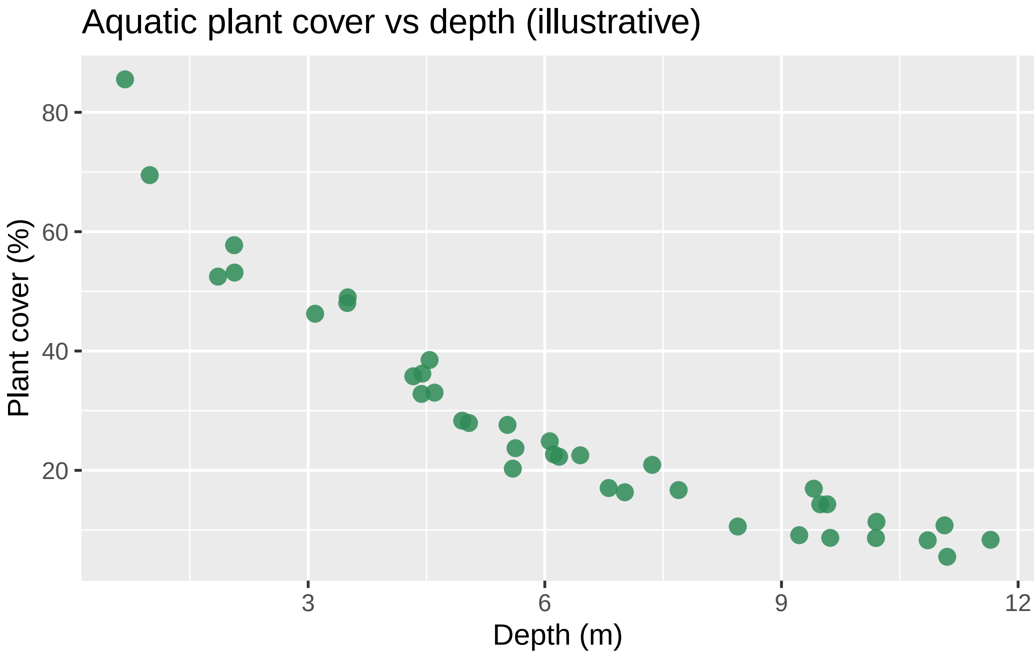
Context: Lake depth (m) vs **submerged macrophyte cover** (%). In deeper water, light drops; cover falls off **quickly** at first, then more slowly (curved decline).

```
set.seed(33)
lake <- tibble(
  depth_m = runif(38, 0.5, 12),
  plant_cover_pct = 90 * exp(-0.22 * depth_m) + rnorm(38, 0, 4)
)

head(lake, 10)
```

```
# A tibble: 10 x 2
   depth_m plant_cover_pct
     <dbl>           <dbl>
 1   5.63             23.7
 2   5.04             27.9
 3   6.06             24.9
 4  11.1              10.8
 5  10.2              11.4
 6   6.45             22.5
 7   5.53             27.6
 8   4.45             36.2
 9   0.678            85.5
10   1.86             52.5
```

```
ggplot(lake, aes(x = depth_m, y = plant_cover_pct)) +
  geom_point(alpha = 0.85, size = 2.5, color = "seagreen") +
  labs(
    title = "Aquatic plant cover vs depth (illustrative)",
    x = "Depth (m)",
    y = "Plant cover (%)"
  )
```



Words: Negative association overall, **strong** trend, **nonlinear** form.

Optional tweaks

- **`alpha`** in **`geom_point()`** helps when points overlap.
- **`size`** adjusts point size for slides or print.
- **`color`** or **`shape`** can encode a third variable later; for **two** numerical variables only, one color for all points is fine.

Summary

For **two numerical** variables, a **scatterplot** in **ggplot2** uses **`aes(x, y)`** and **`geom_point()`**. Describe what you see with **form** (linear vs curved vs none), **direction** (positive, negative, or none), and **strength** (strong through none). Real field and lab data rarely look as clean as teaching simulations, but the same vocabulary organizes your first look before modeling.